CMPE 108 Project Check-In 2
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Predicting How SJSU Students Commute: Analysis of Machine Learning Accuracy and Fairness

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Github Repository: https://github.com/bwallichs/CMPE-108-Project

Abstract (Tentative):
San Jose State University (SJSU) is primarily a commuter campus, and students' transportation needs vary widely due to distance, housing, income, and access to a vehicle or transit pass. This project asks two questions: how accurately can a machine learning classifier predict a student's primary commute mode (driving alone, carpooling, public transportation, walking, biking), and does that accuracy hold up equally across demographic groups? Using SJSU's Annual Student Commute Survey (with the National Household Travel Survey as a backup), we train interpretable models, logistic regression and decision trees, on commute-related features such as distance, housing situation, transportation pass ownership, frequency of travel to campus, and enrollment status. We then evaluate overall performance and group-level performance by age, gender, and income or financial-aid status. Finally, we retrain with sensitive attributes removed and compare overall and per-group accuracy, precision, recall, and F1-score to quantify the accuracy-fairness tradeoff.

Report Outline:

1. Introduction
    a. Background on SJSU as a commuter campus
    b. Problem statement and motivation regarding accuracy and fairness in predictive modeling
2. Related work
    a. Review of SJSU Transportation Solutions and annual commuting statistics
    b. Examination of existing student commute survey data
    c. Machine learning for travel mode prediction
    d. Fairness in travel behavior models
3. Dataset and exploration analysis
    a. Overview of data sources: SJSU Annual Student Commute Survey (response-level data request) and National Household Travel Survey (NHTS)
    b. Feature selection details (commute distance, housing condition/neighborhoods, transit pass ownership, enrollment status)

4. Preprocessing/methods
    a. Pipeline description:

        Receive SJSU Commute Data → Clean & preprocess data → Train ML models ↓
        Test fairness ← Eval. student group performance ← Measure accuracy, precision, etc.

        ↓ Compare accuracy vs. Test fairness → Draw conclusions
    b. Algorithms explored: Logistic Regression and Decision Trees
5. Results
    a. Evaluation Metrics: Accuracy, precision, recall, and F1 score
    b. Analysis of model performance across demographic group attributes (age, gender, income/financial aid)
    c. Impact of removing sensitive features on accuracy-fairness tradeoff(s)
6. Conclusion
    a. Summary of findings, limitations, and future directions

Literature Survey:

After conducting a literature review, prior work demonstrates that tree diagrams do very well in predicting travel mode. The downside to using decision trees is that they are less accurate in representing disadvantaged groups. There are very few sources of studies that focus on the university students specifically, which is the gap we are approaching to fill. Our approach, again, is to test whether removing sensitive features closes group gaps.

- Hagenauer and Helbich (2017) - A Comparative Study of Machine Learning Classifier for Modeling Travel Mode Choice: Random forest was compared with logit and other classifiers for predicting travel mode. It was found that random forests outperformed all of them.
- Zhen et al. (2021) - Equality of Opportunity in travel behavior prediction with deep neural networks and discrete choice models: Analyzed the NHTS dataset and found that there are prediction disparities across demographic groups.
- 2022 Sustainability Survey - Fair Models for Impartial Policies: Controlling Algorithms Bias in Transport Behavioural Modeling: Using mode choice models demonstrated bias related to gender, age, and income of the group.

Approach Pipeline:

Receive SJSU Commute Data → Clean & preprocess data → Train ML models ↓

Test fairness ← Eval. student group performance ← Measure accuracy, precision, etc.

↓ Compare accuracy vs. Test fairness → Draw conclusions

Each stage will become its own script in the repo's src/ folder. That way every model runs through the same data split and metrics.

Algorithms:

1. Logistic Regression: Standard travel mode research.
2. Decision Trees: Show which features separate commute modes.

Repository and Check-In Status:

Github Repository: Active development is tracked under https://github.com/bwallichs/CMPE-108-Project

October 17, 2026: Sent an email to the SJSU Transportation Solutions requesting an anonymized, response-level version of the Annual Student Commute Survey, since the published reports are aggregated and will be an inconvenience when training a classifier.

- Result: No response received yet. Will proceed with the NHTS in the meantime and will switch to SJSU data if permission is granted.

Completed so far:

- Repository structure (data/ and doc/ folders), README, .gitignore, requirements.txt, references, data documentation, check-in log, and project documents.

Next steps:

- Data loading and cleaning scripts, exploratory analysis → logistic regression baseline.

Team Roles:

- Brian Wallichs: Data collection and cleaning; will look at fairness analysis.
- Seline Venzon: Model training and eval; experiments pertaining to accuracy vs. fairness experiments.

Specific Team Contribution:

- Brian Wallichs: Created Abstract, finding sources/references, determine project specifications, determine outline specifications, github commits: CHECK_INS.md, REFERNCES.md, requirements.txt, data/README.md
- Seline Venzon: Determine outline, determine pipeline/approach, finding sources/references, determine project specifications, github commits: README.md, .gitignore, /doc/.gitkeep, doc/CMPE 108 Project Check-In 1.pdf, doc/CMPE 108 Project Proposal.pdf.

References:

1. SJSU Transportation Solutions, Student Commute Survey Report, 2024.
2. Hagenauer, J., & Helbich, M. (2017). A comparative study of machine learning classifiers for modeling travel mode choice. *Expert Systems with Applications*. Link
3. Zheng, Y., Wang, S., & Zhao, J. (2021). Equality of opportunity in travel behavior prediction with deep neural networks and discrete choice models. *Transportation Research Part C*, 132. Link
4. Fair models for impartial policies: Controlling algorithmic bias in transport behavioural modelling. (2022). *Sustainability*, 14(14), 8416. Link